

AUTO REVERSIBLE DC MOTOR With Speed And Rotation Control

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DC motors are widely used in different machines, mechanisms and motion controls in manufacturing and processing industries. In these machines, it is often required to control speed, direction and the number of rotations of the motor. Some applications require only speed control, some only rotation control and some only direction control. Many applications require control of two or all three parameters. Some examples are:

1. To release the thread (or wire) of desired length from a bobbin, it is required to rotate the bobbin for a specific number of rotations. Also, there is a need for controlling the speed at which the thread is released.
2. In a coil-winding machine, it is required to make a specific number of coil turns. This is done by rotating the motor for a specific number of rotations. Also, it is required to vary the motor speed to speed up/down the operation.
3. In a conveyor belt application, to move the object to an exact distance, it is required to rotate the motor for a desired number of rotations.
4. In an automatic storage and retrieval system, it is required to rotate the motor clockwise and anticlockwise to move up and down, or left and right. Also, the motor has to be rotated for a desired number of rotations to reach a specific position.

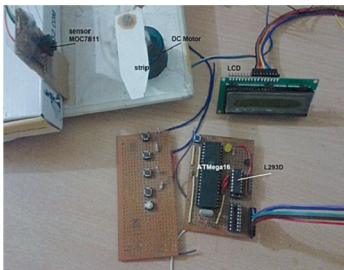


Fig. 1: Author's prototype with motor-sensor arrangement

PARTS LIST

Semiconductors:

- IC1 - MOC7811 opto-interrupter
- IC2 - ATmega16 microcontroller
- IC3 - L293D motor driver
- T1 - 2N222 npn transistor
- LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 330-ohm
- R2, R4-R6 - 1-kilo-ohm
- R3 - 470-ohm
- R7-R11 - 10-kilo-ohm
- R12 - 100-ohm
- VR1 - 1-kilo-ohm preset

Capacitors:

- C1 - 10µF, 16V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- LCD1 - 16×2 LCD module
- M1 - 12V DC motor
- S1-S4, S6 - Tactile switch
- S5 - SPDT switch
- 5V DC and 12V DC regulated power supply
- 40-pin IC base
- Metal strip for attaching on motor shaft

5. In a robotic arm, to pick and place an item, it is required to rotate the motor clockwise and anticlockwise for a desired number of rota-

tions. Speed control is also required for slow or fast operation, accordingly.

There are many more examples and applications where it is required to control speed, direction and rotations of a DC motor.

This project is a demonstration of controlling all three parameters. It is an auto-reversible DC motor that automatically reverses when it completes the desired number of rotations. It is an example of a closed-loop

control system that utilises feedback loop. It counts the actual number of motor rotations and gives feedback to the system. The system compares a set of rotations with actual motor rotations, and when these match, it reverses the motor.

It can repeat this operation in a continuous loop if repeat mode is selected. It also controls the speed of motor from 10 to 100 per cent. The motor starts with 10 per cent speed and gradually attains the selected speed as it completes three to six rotations. Also, when the motor reaches the last six to ten revolutions, the speed gradually decreases, and for the last two rotations it becomes 10 per cent. This speed control is used to precisely stop the motor at the desired number of rotations point and for smooth operation.

The project utilises an ATmega16 microcontroller to control the DC motor, an LCD display to show